Q-TRACE · October 27, 2026 · Noise and mitigation

One-page summary · meeting 4 of 7 · Phases 3B and 4, completed October 7, 2026

Chronology. This meeting presents two completed phases (Oct 7, 2026) in more depth than the Oct 13 overview.

Purpose. To explain what simulated noise does to the frozen fraud classifier, how readout mitigation and zero-noise extrapolation work and what they cost, and what a resource-matched comparison shows.

Key numbers (frozen P3A-R model, validation partition).

Question	Result	Source
Does noise hurt discrimination?	ΔAUPRC ≤ 0.003, every CI contains 0; at least 46 of 47 top frauds kept	`p3b_exact_metrics.csv`
Does noise hurt calibration?	Brier 0.0093 → 0.121; ECE 0.083 → 0.346; precision 0.53 → 0.03	`p3b_exact_metrics.csv`, `p4_exact_diagnostics.csv`
Negative control	symmetric readout: ranking metrics unchanged ($\Delta = 0$)	`p3b_negative_control.csv`
Readout mitigation	Brier 0.053 → 0.011 (high readout noise; 1,024 shots)	`p4_summary.csv`
ZNE	Brier 0.055 → 0.038 (high depolarizing noise)	`p4_summary.csv`
Combined	Brier 0.122 → 0.054 (combined-high); 3× shots: 0.121	`p4_summary.csv`
Costs	ZNE: 3× shots, 9× two-qubit-gate exposure; readout ≈ 2 circuits per batch	`p4_costs.csv`

Takeaways.

- Noise damaged calibration and threshold decisions, not discrimination.
- Mitigation removed a bias that extra shots could not.
- No action improved AUPRC over its resource-matched comparator.
- The best action depended on the noise type, which set up the selection question for Nov 10.

Slide	Content	Speaker (proposed)
1	Title and message	Tushar
2	How we simulated noise	Tushar
3	Calibration, not discrimination	Tushar
4	Two mitigation tools and costs	Mayank
5	Resource-matched comparison	Mayank

Needs confirmation.

- The presenter split.
- The meeting date and format: supplied by the team, not on a public course calendar.